

Curriculum Vitae: Abhishek Dhiman

Personal details

Contact No.: +4915560921957
Date of Birth: 18th July 1996
Location: [BTU Cottbus](#), Germany
Work Permit: EU Blue Card Germany
Email contact: sigmasix09@gmail.com
GitHub Profile: <https://github.com/sigmasix09>
Portfolio Website: <https://sigmasix09.pythonanywhere.com/>
LinkedIn: <https://www.linkedin.com/in/asabhishek09/>
Languages: English (Proficient), Hindi (Native), Germany (Beginner)

Professional Experience

Research Associate (PhD) at Brandenburg Technical University – Rolls Royce **Cottbus, Germany**
Topic: [Virtual Engine Development Using Artificial Intelligence Methods \(VIT-VI\)](#) Nov 2024 to Present

Partners: Rolls-Royce Deutschland (RRD), Brandenburg Technical University (BTU), Technische Hochschule Brandenburg (THB), Friendship Systems (FS), Technical University Berlin (TUB)

- Neural Network (NN) for domain incremental learning on streaming data.
- NN training on large volume, high frequency and high precision numerical data.
- Integration of Virtual Reality (VR) for scientific data visualization of reconstructed data.
- Concurrent simulation and NN training and data compression for storage and VR inference.
- Development of end-to-end seamless pipeline for numeric data compression and reconstruction.
- NN methods for streaming data compression of high precision numerical data, images, videos, etc.
- Supervising Master thesis on topic Machine Learning, Streaming Data Compression at BTU Cottbus.

Master Thesis Student at Siemens Energy AB **Finspång, Sweden**
Topic: [Nearest Neighbour Search Method for 3D Data Mapping of CFD Data](#) Jan 2024 to June 2024

- As part of MSc Aeronautical Engineering at Linköping University.
- Nearest Neighbour Search algorithm development using FORTRAN 95.
- Data mapping of high-resolution data onto low-resolution data for running simulation.
- Algorithm development with compatibility on High Performance Computing (HPC) resources.

Accenture **Bengaluru, India**
Senior Software Engineer: Python Developer (Google Cloud Platform) August 2018 to August 2022

- Software development (backend python developer) and vulnerability fixes.
 - Achievement of more than 50% automation for the corresponding release.
 - Development of REST APIs using FLASK and Python for Data Catalogue.
 - Leading the web automation and scripting for Accenture INTIENT Platform
 - Data management using Graph Database and integration with REST APIs backend.
 - Ontology development on large datasets using RDF format and graph visualization.
 - Enhance REST API response time, team building, SCRUM user story development.
 - Code deployment on GitHub and Google Cloud in development, test and production environment.
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Education

Doctorate: Brandenburg Technical University VIT VI - Virtual Engine Development with Artificial Intelligence Methods Topic: Concurrent Data Compression & Reconstruction using Neural Networks for VR Visualization	Cottbus, Germany Nov 2024 to Present
Master of Science in Aeronautical Engineering (2 years, 120 credits) Linköping University, Sweden (abh352@student.liu.se), Master's degree	Linköping, Sweden August 2022 to June 2024
Bachelor of Aeronautical Engineering, Bachelor's degree Dayananda Sagar College of Engineering [Grade: 74.30 % CGPA: 7.82 / 10.0] Thesis: " Shockwave Boundary Layer Interaction & Assessment of Passive Flow Control Device ".	Bengaluru, India August 2014 to July 2018

Professional Skills

Technical Skills: Incremental Machine Learning, Python, CFD, MATLAB, FORTRAN95, PyTorch, Neural Networks, Data Compression, Hybrid InSitu InTransit approach, HPC, Analytical and Statistical skills, Data Science, Software development, SCRUM, Backend Development, GraphDB, Python-Flask, Rest APIs, Scientific Data Visualization, Virtual reality (VR), Scientific Report Writing

Softwares and Frameworks: PyTorch, Ansys Fluent, ParaView Kitware, MATLAB, ABAQUS, GraphDB, VSCode, GCP and IBM cloud, Jupyter Lab, GitHub, JIRA Atlassian, Latex Overleaf

Certifications

Digitalization in Aeronautics and Space: Professional Certificate (Verified) Authorized by Technical University of Munich, COURSERA	Bengaluru, India July 2021 to September 2021
IBM Data Science: Professional Certificate, Duration: 11 months (Verified) Project: https://github.com/sigmasix09/Coursera_Capstone	Bengaluru, India April 2021 to July 2021
ENGR2000X: A Hands-on Introduction to Engineering Simulations (Verified) Developed by Cornell University, EDX	Bengaluru, India November 2020 to February 2021
Mastering Ansys CFD (Level 1), Udemy. (Verified) Ansys Fluent, ICEM CFD, CFX, Tech Plot. Duration (20 hours)	Bengaluru, India August 2020 to October 2020
Probability and Statistics in Data Science using Python (Verified) Prepared by UCSanDiegoX, EDX. Duration (120 hours)	Bengaluru, India May 2020 to August 2020
Python Developer: Zero to Mastery, Udemy. (Verified) Conducted by Andrei Neagoie. Duration (30.5 hours)	Bengaluru, India January 2020 to May 2020
Introduction to Programming with MATLAB, Coursera. (Verified) Offered by Vanderbilt University, Nashville. Duration (35 hours)	Bengaluru, India November 2019 to February 2020

Hobbies and Interests

Badminton, Cooking, Long walks, Watching Anime